1 ER-DRO model (with CVaR-type ambiguity set)

Data: (y^s, x^s), $s \in S$—for example, city-year observations of choice probabilities y^s and city characteristics x^s. Each observation corresponds to one scenario, so S is the set of observation indices. Regression gives $\hat{f}$ and residuals $\hat{\varepsilon}^s = y^s - \hat{f}(x^s)$ (4.A); the characteristics x_c of candidate city c are used to generate scenarios D_c^s (4.B).

$$\min_{y,q,O,U,\tau,w} \quad \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle costs}} + \tau + \underbrace{\frac{1}{|S|(1-\zeta)} \sum_{s \in S} w^s}_{\text{CVaR of scenario penalties}}$$

Note. The baseline term $\frac{1}{|S|}\sum_s H(q, D^s)$ is replaced by $\tau + \frac{1}{|S|(1-\zeta)}\sum_s w^s$, where $H(q, D^s) = c_c^+ O_c^s + c_c^- U_c^s$. Context x_c enters only the scenario values $D_c^s = M_c p_c^s$; the optimization structure remains unchanged.

2 Constraints

$$\begin{aligned}
w^s &\geq \sum_c \left(c_c^+ O_c^s + c_c^- U_c^s\right) - \tau && \forall s \in S, \\
\kappa_c q_c - D_c^s &= O_c^s - U_c^s && \forall c \in C,\ s \in S, \\
0 \leq q_c &\leq \bar{q}_c y_c && \forall c \in C, \\
\sum_{c \in C} q_c &\leq Q, \\
y_c \in \{0,1\}, \quad q_c &\in \mathbb{Z}_{\geq 0} && \forall c \in C, \\
O_c^s, U_c^s &\geq 0 && \forall c \in C,\ s \in S, \\
w^s &\geq 0 && \forall s \in S, \qquad \tau \in \mathbb{R}.
\end{aligned}$$

Constraint	Meaning
$0 \leq q_c \leq \bar{q}_c y_c$	Zero vehicles if the city is not entered; up to the city-specific limit if it is entered
$\sum_c q_c \leq Q$	Total vehicle limit
$\kappa_c q_c - D_c^s = O_c^s - U_c^s$	The difference between service capacity and robotaxi demand equals unused capacity minus unmet demand.
$w^s \geq \sum_c (c_c^+ O_c^s + c_c^- U_c^s) - \tau$, $w^s \geq 0$	Excess of scenario s's penalty over threshold τ. Optimization makes w^s as small as possible, so $w^s = (\text{penalty}_s - \tau)^+$.

3 Notation

3.1 Contextual + Additional DRO Notation

Symbol	Meaning	Type and unit
(y^s, x^s)	Dependent variable (choice probability) and covariates of observation s	Data; $s \in S$
x_c	Characteristic (covariate) vector of candidate city c	Input; corresponds to Kannan's new observation x
$\hat{f}$	Estimated regression function; $\hat{f}(x) \approx \mathbb{E}[y \mid x]$	Estimated function
$\hat{\varepsilon}^s$	Residual $y^s - \hat{f}(x^s)$	Derived; on the p scale
ζ	Radius (tail proportion) of the CVaR-type ambiguity set. $1-\zeta$ is the share of scenarios retained as the worst group	Input; $[0, 1)$
τ	Penalty threshold. A decision variable; at the optimum, the "$\lvert S\rvert(1-\zeta)$-th largest penalty"	Decision variable; currency
w^s	Amount by which scenario s's penalty exceeds τ	Decision variable; currency

3.2 Baseline Notation

Symbol	Meaning	Type and unit
C	Set of potential candidate cities	Input set; cities
S	Set of demand scenarios	Input set; scenarios. Here it is the set of regression-data observation indices; $\|S\|$ = number of observations
y_c	1 if city c is entered, else 0	Decision variable; binary (0 or 1)
q_c	Number of vehicles deployed in city c	Decision variable; vehicles (integer)
κ_c	Number of trips one vehicle can serve during the period	Scale parameter; trips/vehicle over the evaluation period
Q	Total number of vehicles available for deployment	Input parameter; vehicles
$\bar{q}_c$	Upper bound on vehicle deployment in each city	Input parameter; vehicles
F_c	Fixed cost of city entry and basic infrastructure	Input parameter; currency over the evaluation period
K_c	Per-vehicle preparation, deployment, and holding cost over the evaluation period	Input parameter; currency/vehicle over the evaluation period
c_c^+	Additional loss per unit of unused service capacity	Input parameter; currency/trip of unused capacity
c_c^-	Foregone contribution margin per unit of unserved potential demand	Input parameter; currency/unserved trip
M_c	Total number of **potential travel opportunities across all modes** in city c during one year	Input parameter; usage opportunities over the evaluation period. Data or assumption
p_c^s	Probability of choosing robotaxi service for a travel opportunity in city c	Derived input (generated by regression); dimensionless probability in $[0,1]$. $p_c^s = \text{proj}_{[0,1]}(\hat{f}(x_c) + \hat{\varepsilon}^s)$
D_c^s	Robotaxi choice demand	Derived input; trips over the evaluation period. $D_c^s = M_c p_c^s$
O_c^s	Unused service capacity	Outcome variable; trips over the evaluation period. $O_c^s = (\kappa_c q_c - D_c^s)^+$
U_c^s	Unserved robotaxi demand	Outcome variable; trips over the evaluation period. $U_c^s = (D_c^s - \kappa_c q_c)^+$

4 Explanations

4.1 Derivation of the CVaR Formulation with Contextual Demand

A. Definitions.

- $H(q, D^s)$: scenario s's second-stage cost ("Common second stage" at the top of the document).

$$H(q, D^s) = \min_{O^s, U^s \geq 0} \sum_c (c_c^+ O_c^s + c_c^- U_c^s)$$
$$\text{s.t.} \quad \kappa_c q_c - D_c^s = O_c^s - U_c^s \quad \forall c.$$

  At the optimum, $O_c^s = (\kappa_c q_c - D_c^s)^+$ and $U_c^s = (D_c^s - \kappa_c q_c)^+$, so

$$H(q, D^s) = \sum_c \left[c_c^+ (\kappa_c q_c - D_c^s)^+ + c_c^- (D_c^s - \kappa_c q_c)^+ \right].$$

- w^s: the amount by which scenario s's penalty exceeds threshold τ. At the optimum, $w^s = (H(q, D^s) - \tau)^+$.
- $D_c^s(x_c) = D_c^s = M_c p_c^s = M_c \, \text{proj}_{[0,1]}(\hat{f}(x_c) + \hat{\varepsilon}^s), \quad D^s = (D_1^s, D_2^s, \ldots, D_{|C|}^s)$.
-

$$\hat{P}_n^{\text{ER}}(x) = \frac{1}{|S|} \sum_{s \in S} \delta_{D^s(x)}, \qquad \hat{\mathcal{P}}_n(x) = \left\{ \sum_s p_s \delta_{D^s(x)} : p \in \mathcal{P}_n(x; \zeta) \right\}.$$

B. Chain of equalities. Starting from Kannan's residuals-based DRO formulation

$$\min_{z \in Z} \left[\sup_{Q \in \hat{\mathcal{P}}_n(x)} \mathbb{E}_{Y \sim Q}[c(z, Y)] \right],$$

fix the decision $z = (y, q)$ and expand the inner supremum first.

$$\sup_{Q \in \hat{\mathcal{P}}_n(x)} \mathbb{E}_{Y \sim Q}[c(z, Y)]$$

$$
\begin{aligned}
&\overset{(a)}{=} \sup_{p\in\mathcal{P}_n(x;\zeta)} \sum_{s\in S} p_s c(z, D^s(x)) \\
&\overset{(b)}{=} \sum_c F_c y_c + \sum_c K_c q_c + \sup_{p\in\mathcal{P}_n} \sum_{s\in S} p_s H(q, D^s) \\
&\overset{(c)}{=} \sum_c F_c y_c + \sum_c K_c q_c + \min_{\tau\in\mathbb{R}} \left[\tau + \frac{1}{|S|(1-\zeta)} \sum_{s\in S} (H(q, D^s) - \tau)^+ \right] \\
&\overset{(d)}{=} \sum_c F_c y_c + \sum_c K_c q_c + \min_{\tau, w} \left\{ \begin{array}{l} \tau + \dfrac{1}{|S|(1-\zeta)} \displaystyle\sum_{s\in S} w^s : \\ w^s \geq H(q, D^s) - \tau, \quad w^s \geq 0 \quad \forall s \end{array} \right\} \\
&\overset{(e)}{=} \sum_c F_c y_c + \sum_c K_c q_c + \min_{\tau, w, O, U} \left\{ \begin{array}{l} \tau + \dfrac{1}{|S|(1-\zeta)} \displaystyle\sum_{s\in S} w^s : \\ w^s \geq \displaystyle\sum_c (c_c^+ O_c^s + c_c^- U_c^s) - \tau, \\ \kappa_c q_c - D_c^s = O_c^s - U_c^s, \\ w^s, O_c^s, U_c^s \geq 0 \quad \forall c, s \end{array} \right\}.
\end{aligned}
$$

Finally, apply the outer $\min_{y,q}$. Since $\min_{y,q} \min_{\tau,w,O,U} = \min_{y,q,O,U,\tau,w}$, the result is

$$
\begin{aligned}
\text{(f)} \quad \min_{y,q,O,U,\tau,w} \quad & \sum_c F_c y_c + \sum_c K_c q_c + \tau + \frac{1}{|S|(1-\zeta)} \sum_{s\in S} w^s \\
\text{s.t.} \quad & w^s \geq \sum_c (c_c^+ O_c^s + c_c^- U_c^s) - \tau, \quad w^s \geq 0 \quad \forall s \in S, \\
& \kappa_c q_c - D_c^s = O_c^s - U_c^s, \quad O_c^s, U_c^s \geq 0 \quad \forall c \in C,\ s \in S, \\
& 0 \leq q_c \leq \bar{q}_c y_c, \quad \sum_c q_c \leq Q, \\
& y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\geq 0}, \quad \tau \in \mathbb{R}.
\end{aligned}
$$

C. Why each equality holds.

- (a) The ambiguity set has the same support, so $Q = \sum_s p_s \delta_{D^s}$ and expectation is a weighted sum.
- (b) In $c(z, D^s) = \sum_c F_c y_c + \sum_c K_c q_c + H(q, D^s)$, the first two terms do not depend on s. Since $\sum_s p_s = 1$, they can be taken outside the supremum.
- (c) If $\mathcal{P}_n$ is the box in Example 1, $p_s \leq 1/(|S|(1-\zeta))$, the supremum is the average of the $|S|(1-\zeta)$ largest penalties, namely CVaR_ζ. This is equivalent to the Rockafellar–Uryasev threshold representation.
- (d) Linearization of $(\cdot)^+$. Minimizing w^s under $w^s \geq H - \tau$ and $w^s \geq 0$ gives $w^s = (H-\tau)^+$, so the values agree.
- (e) $H(q, D^s)$ is itself a minimum over O^s, U^s. "$w^s \geq \min_{O,U}(\cdots) - \tau$" is equivalent to "$w^s \geq (\cdots) - \tau$ for some feasible O^s, U^s." The inner minimum can therefore be merged into the outer minimum and the balance equation imposed as a constraint.
- (f) The outer $\min_{y,q}$ and inner $\min_{\tau,w,O,U}$ are both minimizations and can be combined. The supremum has already become a minimum in (c), so no min–max structure remains.

4.2 Scenario Generation

A. Regression

For the characteristics x_c of candidate city c,

$$
p_c^s = \mathrm{proj}_{[0,1]}(\hat{f}(x_c) + \hat{\varepsilon}^s), \qquad D_c^s = M_c p_c^s, \qquad s \in S.
$$

Fit $\hat{f}$ and calculate residuals $\hat{\varepsilon}^s = y^s - \hat{f}(x^s)$, $s \in S$. x^s are training points. x and ϵ are assumed to follow $N(0,1)$ and to be independent of the covariate x (homoscedasticity must be checked after regression). As in the baseline model, M_c is held fixed.

B. What are the covariates X?

B-1. How is M determined? For details, see `2.driving factors/C.Selection/Factors_of_M`.

M_c is constructed from NHTS group-specific per-person trip rates × city populations. This gives a fixed value calculated for all candidate cities using the same definition, without regression.

B-2. How is p determined?

For details, see `2.driving factors/C.Selection/Factors_of_p`.

The preliminary selection identifies two city characteristics and three decision-dependent service variables as initial priorities. Test the two exogenous variables in the current model first; endogenous service variables will be incorporated into a decision-dependent ER-DRO extension. Exogenous/endogenous refers to dependence on operating decisions.

- **Exogenous:** zero-vehicle household share (exogenous), age composition (exogenous).
- **Endogenous:** relative fare (endogenous), relative waiting time (endogenous), in-vehicle support features (endogenous).

C. Which regression method will be used, and why? Still to be decided

Note. Kannan et al. (Residuals-based Distributionally Robust Optimization)

Can any regression method be used? The paper's framework is method-agnostic (OLS, Lasso, Ridge, kNN, and random forests are all possible). Three considerations nevertheless matter.

- **Theoretical conditions.** Assumptions 2, 8, and 15 concern how quickly regression error decreases with $|S|$. Parametric methods such as OLS and Lasso satisfy $r = 1$ (error $\sim |S|^{-1/2}$); nonparametric methods are slower, with $r = O(1)/d_x$. With few cities and many features, parametric methods are needed for the theory to apply.
- **Regularize with small samples.** In the paper's experiments, Lasso and Ridge outperformed OLS when $|S|$ was small relative to d_x and complemented DRO. Small $|S|$ also makes training residuals optimistically small, so leave-one-out (jackknife) residuals from the previous paper [34] are preferable.
- **Model-structure assumption.** In $Y = f^*(X) + \epsilon$, errors must be independent of X (homoscedastic) for residuals to be shared across cities. Regressing trip counts D violates this assumption because larger cities tend to have larger errors. Regressing choice probability p (or logit p) on city characteristics and constructing $D_c = M_c p_c$ is much closer to homoscedasticity; projection onto $[0, 1]$ is also natural. If heteroscedasticity remains, use the authors' subsequent extension [35].

D. Selecting the Ambiguity Set

We use the CVaR-based ambiguity sets with the same support as $\hat{P}_n^{\mathrm{ER}}$, the empirical distribution of the residuals.

Note. Kannan et al. (Residuals-based Distributionally Robust Optimization)

Ambiguity sets with the same support as $\hat{P}_n^{\mathrm{ER}}(x)$ (cf. [4, 6], for instance): given radius $\zeta_n(x) \geq 0$, set

$$\hat{\mathcal{P}}_n(x) = \left\{ Q = \sum_{i=1}^{n} p_i \delta_{\mathrm{proj}_Y(\hat{f}_n(x)+\hat{\epsilon}_n^i)} : p \in \mathcal{P}_n(x; \zeta_n(x)) \right\},$$

where $\mathcal{P}_n(x; \zeta_n(x))$ is a generic ambiguity set for the n-dimensional vector of probabilities p. We focus on sets $\mathcal{P}_n(x; \zeta_n(x))$ that satisfy for each $x \in X$

$$p \in \mathbb{R}_+^n \quad \text{and} \quad \sum_{i=1}^{n} p_i = 1, \quad \forall p \in \mathcal{P}_n(x; \zeta_n(x)),$$

$$\lim_{\zeta \downarrow 0} \mathcal{P}_n(x; \zeta) = \mathcal{P}_n(x; 0) = \left\{ \left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n} \right) \right\}. \tag{8}$$

Example 1. CVaR-based ambiguity set [45, 48]: given radius $\zeta_n(x) \in [0, 1)$, set

$$\mathcal{P}_n(x; \zeta_n(x)) := \left\{ p \in \mathbb{R}_+^n : \sum_{i=1}^{n} p_i = 1, \quad p_i \leq \frac{1}{n(1 - \zeta_n(x))}, \quad \forall i \in [n] \right\}.$$

Observe that $\zeta_n(x)$ enters the ambiguity set $\mathcal{P}_n(x; \zeta_n(x))$ through the CVaR risk parameter.

D-1. Why choose CVaR?

If we do not fully trust assigning probability $1/|S|$ to each of the $|S|$ scenarios, two things may be questioned. Doubting scenario values leads to sets that move points (Wasserstein, SRO); doubting scenario probabilities leads to sets that change only weights (CVaR-type). We choose the latter.

- **The structure is preserved.** The "average of the $(1-\zeta)$ fraction of scenarios with the largest penalties" can be added directly to the existing MILP using only threshold τ and excess variables w^s. Point-moving sets introduce min–max–min within integer decisions; the paper also regards this as generally difficult.
- **Point-moving sets have different meanings depending on the c^+/c^- ratio.** If $c^+ \ll c^-$, the worst-case distribution pushes demand only upward, making the experiment effectively similar to demand-multiplier sensitivity. As c^+ approaches c^-, it also pushes toward unused capacity, complicating interpretation. CVaR-type sets ask the same question regardless of the ratio: "What if bad scenarios occur more frequently?" This makes comparisons across ratio combinations clearer.
- **Frequency has the weakest basis, rather than values.** ER-DRO scenario values are grounded in data—regression predictions + actual residuals—but with fewer observations, assigning exactly $1/|S|$ to every scenario is the most fragile assumption. A probability-only ambiguity set targets that weakness.
- **Easy to explain.** $\zeta = 0.5$ means "minimize the average of the worst half." Results can be read through the "value of one vehicle" logic by replacing scenario proportions with worst-case weights.

Caveat: in the paper's experiments, the Hellinger-type set from the same class yielded smaller improvements than Wasserstein. Those small-sample continuous-distribution settings differ from ours, but validation is needed; run both sets and provide a comparison table.

D-2. Understanding the "τ trick"

Note. Example (the numbers below are from the v3 instance without covariates, i.e., ten manually designed scenarios).

Fix the reference allocation at $(132, 368, 500)$ and set $\zeta = 0.5$ (worst five scenarios). Scenario penalties $C_s = \sum_c c_c^- U_c^s$ and the solver's τ, w^s are as follows (million USD).

s	8	4	5	6	10	9	1	3	2	7
C_s	297.7	288.0	220.6	174.4	156.5	114.7	104.0	58.8	56.9	6.3
$w^s = (C_s - \tau)^+$	141.2	131.5	64.1	17.9	0	0	0	0	0	0

$\tau = 156.5$ (the fifth-largest penalty), and the objective is

$$18.47 + 156.5 + \frac{1}{5}(141.2 + 131.5 + 64.1 + 17.9) = 245.9.$$

This is exactly fixed costs + the worst-five average, 227.4. Thus w^s is "the amount by which scenario s's penalty exceeds the threshold," and is zero outside the worst group. Changing the allocation changes C_s, and τ, w^s change accordingly.

I do not set τ. Like y, q, τ is a decision variable found by the solver. At the optimum it is always the "$|S|(1-\zeta)$-th largest penalty." I set only ζ (using one of the three methods in 4.D).

Note. References

- Rockafellar & Uryasev (2000), "Optimization of conditional value-at-risk," *Journal of Risk*. Origin of the τ trick. They use α for the threshold and β for the tail level (our τ, ζ).
- Shapiro, Dentcheva, Ruszczyński, *Lectures on Stochastic Programming*, chapter on risk-averse optimization. The equivalence of CVaR's "$\min_\tau$ threshold representation" and "worst-case weights with probabilities capped at $1/(|S|(1-\zeta))$" is presented there. The paper's Example 1 uses the latter directly.
- Section 7 of Kannan's paper: the portfolio objective is written as $\min_\tau \mathbb{E}[\tau + \frac{1}{1-\beta}\max\{0, \cdot - \tau\}]$ and solved as an LP. This is the same approach as ours.

E. Methods for Selecting ζ (Radius)

In the CVaR-type set, ζ is not a distance radius but a tail proportion determining "how many worst scenarios to consider" ($1 - \zeta$ = retained proportion).

E-1. Cross validation

E-2. Intuitively preset ζ according to the decision-maker's (researcher's) desired level of risk aversion (Risk Preference)

Note. Kannan et al. (Residuals-based Distributionally Robust Optimization)

Algorithm 2 Specifying a covariate-independent radius ζ_n using the ER-DRO problem

Input: data $\mathcal{D}_n$, set of candidate radii Δ, number of folds K, and number of covariate realizations sampled during each fold $T \leq \lfloor n/K \rfloor$.
Partition $[n]$ into subsets $S_1, \ldots, S_K$ of (roughly) equal size at random. Let $\mathcal{D}_{-k} := \mathcal{D}_n \setminus \{(y^i, x^i)\}_{i \in S_k}$.
for $k = 1, \ldots, K$ **do**
  Pick without replacement a random subset $\bar{X}$ of $\{x^i\}_{i \in S_k}$ of size T.
  for $\bar{x} \in \bar{X}$ **do**
    for $\zeta \in \Delta$ **do**
      Fit a regression model $\hat{f}_{-k}$ using the data $\mathcal{D}_{-k}$ and compute its in-sample residuals $\{\hat{\varepsilon}^i_{-k}\}_{i \notin S_k} := \{y^i - \hat{f}_{-k}(x^i)\}_{i \notin S_k}$.
      Solve the ER-DRO problem below at covariate $\bar{x}$ to get solution $\hat{z}^{\mathrm{DRO}}_{-k}(\bar{x}, \zeta)$

$$\min_{z \in Z} \sup_{Q \in \hat{\mathcal{P}}_{-k}(\bar{x})} \mathbb{E}_{Y \sim Q}[c(z, Y)],$$

where the ambiguity set $\hat{\mathcal{P}}_{-k}(\bar{x})$ with radius ζ is centered at the estimated empirical distribution

$$\hat{P}^{\mathrm{ER}}_{-k}(\bar{x}) := \frac{1}{n - |S_k|} \sum_{i \notin S_k} \delta_{\hat{f}_{-k}(\bar{x}) + \hat{\varepsilon}^i_{-k}}.$$

    end for
  end for
end for
Output: Radius

$$\zeta_n \in \arg\min_{\zeta \in \Delta} \frac{1}{T} \sum_{\bar{x} \in \bar{X}} \frac{1}{K} \sum_{k \in [K]} \frac{1}{|S_k|} \sum_{i \in S_k} c\Big(\hat{z}^{\mathrm{DRO}}_{-k}(\bar{x}, \zeta), y^i\Big)$$

for the ambiguity set $\hat{\mathcal{P}}_n(x)$ for the ER-DRO problem (7).

Algorithm 3 Specifying a covariate-dependent radius $\zeta_n(x)$ using the ER-DRO problem

Input: data $\mathcal{D}_n$, set of candidate radii Δ, number of folds K, and new covariate realization $x \in X$.
Partition $[n]$ into subsets $S_1, \ldots, S_K$ of (roughly) equal size at random. Let $\mathcal{D}_{-k} := \mathcal{D}_n \setminus \{(y^i, x^i)\}_{i \in S_k}$.
for $k = 1, \ldots, K$ **do**
  for $\zeta \in \Delta$ **do**
    Fit a regression model $\hat{f}_{-k}$ using the data $\mathcal{D}_{-k}$ and compute its in-sample residuals $\{\hat{\varepsilon}^i_{-k}\}_{i \notin S_k} := \{y^i - \hat{f}_{-k}(x^i)\}_{i \notin S_k}$.
    Solve the ER-DRO problem below at covariate x to obtain solution $\hat{z}^{\mathrm{DRO}}_{-k}(x, \zeta)$

$$\min_{z \in Z} \sup_{Q \in \hat{\mathcal{P}}_{-k}(x)} \mathbb{E}_{Y \sim Q}[c(z, Y)],$$

where the ambiguity set $\hat{\mathcal{P}}_{-k}(x)$ with radius ζ is centered at the estimated empirical distribution

$$\hat{P}^{\mathrm{ER}}_{-k}(x) := \frac{1}{n - |S_k|} \sum_{i \notin S_k} \delta_{\hat{f}_{-k}(x) + \hat{\varepsilon}^i_{-k}}.$$

    Fit a regression model $\hat{f}_k$ using the data $\{(y^i, x^i)\}_{i \in S_k}$ and compute its in-sample residuals $\{\hat{\varepsilon}^i_k\}_{i \in S_k} := \{y^i - \hat{f}_k(x^i)\}_{i \in S_k}$.
  end for
end for
Output: Radius

$$\zeta_n(x) \in \arg\min_{\zeta \in \Delta} \frac{1}{K} \sum_{k \in [K]} \frac{1}{|S_k|} \sum_{i \in S_k} c\Big(\hat{z}^{\mathrm{DRO}}_{-k}(x, \zeta), \hat{f}_k(x) + \hat{\varepsilon}^i_k\Big)$$

for the ambiguity set $\hat{\mathcal{P}}_n(x)$ for the ER-DRO problem (7).